

USMAN GANI PRANTO

Undergraduate Researcher in Mechanical Engineering, Bangladesh University of Engineering and Technology (BUET)

Dhaka, Bangladesh | praanto17@gmail.com | praaaanto.github.io

linkedin.com/in/praaaanto | github.com/praaaanto | ORCID [0009-0005-6870-2118](https://orcid.org/0009-0005-6870-2118)

Research Interests

Atomistic and multiscale modelling of materials and interfaces; thermomechanical reliability of electronic packaging; phase change heat transfer and thermal energy storage; computational heat transfer and fluid flow.

Education

Bangladesh University of Engineering and Technology (BUET) Nov 2022 – Jul 2027 (expected)

B.Sc. in Mechanical Engineering | CGPA 3.57/4.00 after six of eight terms

Dhaka, Bangladesh

- **Thesis:** *Atomistic Molecular Dynamics of Thermomechanical Reliability in Copper/SAC305/Copper Solder Joint Stacks*. Supervisor: Dr. M. A. Motalab. Pre-defence held 30 Sep 2026; final defence expected Apr 2027.
- **Selected coursework:** Heat Transfer; Thermodynamics; Fluid Mechanics; Solid Mechanics; Numerical Analysis; Machine Design; Heat Transfer Equipment Design (ME 310, capstone).

Notre Dame College

Jun 2019 – Dec 2021

Higher Secondary Certificate (HSC), Science | GPA 5.00/5.00; ranked 25th of 2,000+ students

Dhaka, Bangladesh

Mymensingh Zilla School

Feb 2019

Secondary School Certificate (SSC), Science | GPA 5.00/5.00

Mymensingh, Bangladesh

Publications and Conference Papers

* denotes corresponding author. Each entry states its exact current status.

- [C1] S. M. S. A. Sohan, **U. G. Pranto***, M. F. Faisal, M. S. H. Kafi, and K. M. Ar-Rafi, “Design and Development of a Honeycomb-Based Latent Heat Thermal Energy Storage (LHTES) Unit Using Paraffin PCM for Solar-Assisted Domestic Hot Water Systems,” *ASHRAE 2027 Winter Conference*, Chicago, IL, USA, Jan 23–27, 2027. **Abstract accepted; full manuscript under review** (submitted Sep 2026).
- [C2] **U. G. Pranto** and M. A. Motalab, “Monotonic and Cyclic Shear of SAC305 Solder with Discrete Intermetallic Precipitates: A Molecular Dynamics Study,” *10th BSME International Conference on Thermo-Fluid Engineering (ICTE 2026)*, BUET, Dhaka, Dec 19–21, 2026. **Abstract accepted; full paper submitted** (Sep 2026).

Research Experience

Undergraduate Thesis Researcher, Atomistic Simulation of Solder Joints Jul 2026 – Present

Supervisor: Dr. Mohammad Abdul Motalab, Department of Mechanical Engineering

BUET, Dhaka

- Investigating, through molecular dynamics (MD) in LAMMPS with second nearest neighbour modified embedded atom method (2NN-MEAM) potentials, how intermetallic microstructure governs the shear and fatigue response of Cu/SAC305/Cu joint stacks, extending to explicit Cu₃Sn and Cu₆Sn₅ intermetallic compound (IMC) layers.
- Introduced a microstructure-resolved SAC305 model that builds Ag₃Sn and Cu₆Sn₅ as discrete precipitates, in place of the random Ag and Cu substitution used in prior atomistic studies, isolating the effect of spatial arrangement at fixed composition.
- Validated the simulation pipeline by reproducing a published tensile benchmark (Fardin *et al.*, *J. Electron. Mater.*, 2025) to within 2% in ultimate tensile strength and Young’s modulus.
- Verified the Ag–Cu–Sn 2NN-MEAM potential (Ko, Lee and Kim, *J. Mater. Res.*, 2022) against reference lattice constants and formation energies of the elemental and intermetallic phases before production runs.
- Built and profiled a 14-rank MPI (Message Passing Interface) production workflow on an institutional virtual machine; load balancing cut inter-rank workload imbalance from 367% to 16% and sustained 7.4 ns/day.
- Characterising plasticity and defect evolution in OVITO using centrosymmetry (CSP), dislocation extraction (DXA), and common neighbour analysis (CNA).

Researcher, Latent Heat Thermal Energy Storage (LHTES)

Oct 2025 – Present

Supervisor: Saifuddin Zafar, Lecturer, Department of Mechanical Engineering

BUET, Dhaka

- Designed a honeycomb-encapsulated LHTES unit with 19 hexagonal cells of Rubitherm RT50 paraffin phase change material (PCM) for solar-assisted domestic hot water, developed from the ME 310 capstone into a conference paper.
- Led fabrication of the prototype (352.8 kJ design storage capacity) and its 100 psi pressure test.

- Performed transient computational fluid dynamics (CFD) in ANSYS Fluent with the enthalpy–porosity formulation to resolve melting and solidification fronts, and evaluated charging and discharging performance against cell geometry.
- Wrote the manuscript as corresponding author and prepared the poster and presentation [C1].

Research Grant Proposals

RISE Student Research Grant, BUET

Sep 2026

- Proposal to the Research and Innovation Centre for Science and Engineering (RISE): *Atomistic Investigation of Thermomechanical Reliability in Cu/SAC305/Cu Multi-Layer Nanoelectronic Package Stacks*. BDT 99,775 requested for Jul 2026 – Jun 2027. **Submitted; under review, not yet awarded.**

Presentations and Talks

- Undergraduate thesis pre-defence, Department of Mechanical Engineering, BUET, Dhaka. **30 Sep 2026**
- *Before You Apply*: guidance for ASHRAE scholarship applicants (co-presenter), ASHRAE Beyond Membership webinar, ASHRAE Bangladesh Chapter, online. **29 Aug 2026**

Honors and Awards

- **ASHRAE Scholarship Trustees General Scholarship** (2026–27): USD 1,500; among the first recipients from Bangladesh in ASHRAE Region-at-Large.
- **BUET Technical Scholarship**: merit scholarship for the 2021–22 and 2022–23 academic sessions.
- **4th Place, Intra BUET Robo Challenge** (2024): SoccerBOT, competing against university teams.
- **Government Talentpool Scholarship**: national merit scholarship awarded at the PSC, JSC, and SSC public examinations.

Technical Skills

Atomistic simulation: LAMMPS (2NN-MEAM and EAM potentials; NVT and NPT ensembles), OVITO (CSP, DXA, CNA), ASE, Materials Project, CIF-to-LAMMPS structure generation

Continuum simulation: ANSYS Fluent (transient melting and solidification), ANSYS Mechanical

High-performance computing: MPI (MPICH, MS-MPI), Linux, WSL2, remote virtual machine job management

Programming: Python (NumPy, pandas, Matplotlib), C, C++, MATLAB

Design and tools: SOLIDWORKS (CSWP), AutoCAD, Git, L^AT_EX and Overleaf, Jupyter, Arduino

Professional Experience

Meghna Sugar Refineries Ltd.

Jun 2026

Industrial Attachment Trainee

Meghnaghat, Narayanganj, Bangladesh

- Studied the thermal and mechanical systems of a high-capacity refining line: ion-exchange resin decolorisation, multi-effect evaporation, vacuum crystallisation, and a 32 MW co-generation plant.
- Reviewed preventive maintenance of high-speed centrifuges and fluidised bed dryers; authored 14 technical reports on SCADA process control, machine-shop fabrication, and water and effluent treatment utilities.

Engineering Projects and Design Teams

Oogway, Bio-Mimetic Sea Turtle Robot | SOLIDWORKS, Arduino, sensors

2025

- Engineered the shell, flippers, and sealed motor pods of an underwater robot with flapping-fin propulsion and dynamic buoyancy control; verified waterproofing and hydrodynamic performance in reservoir trials.

SoccerBOT, Remote-Controlled Mecanum-Wheel Robot | SOLIDWORKS, AutoCAD, PCB

2024 – 2025

- Led chassis design and mechanical assembly and integrated drive hardware with microcontroller interfaces; placed 4th at the Intra BUET Robo Challenge 2024 and competed in the Soccer Bot segment of UAP Techtron 2025, University of Asia Pacific.

Team Interplanetar, BUET Mars Rover Team

Aug 2025 – Jul 2026

Team member, Mechanical and Manufacturing Subteam

BUET, Dhaka

- Contributed to wheel design, an Ackermann steering system, and robotic gripper assembly, with design documentation.

Leadership and Service

ASHRAE BUET Student Branch

Oct 2024 – Present

President (Jul 2026 – Present); Manager (May 2025 – Jul 2026); founding member

BUET, Dhaka

- Constituted and onboarded the 20-member 3rd Executive Committee and reconciled branch records for submission to ASHRAE.
- Organised the branch visit of Mr. Abdallah Mahmoud, Regional Vice-Chair for Government Affairs, ASHRAE Region-at-Large (Sep 2026), hosting about 15 senior ASHRAE Bangladesh Chapter leaders and 30+ student members.
- Coordinating BUET delegation logistics for the ASHRAE Region-at-Large Chapters Regional Conference (CRC), Cairo, Oct 2026.
- As Manager: exhibited three mechanical prototypes, including heat exchangers and a cooling tower, at SAFECON 2025 and ICME 2025; represented the ASHRAE Bangladesh Chapter at the 10th Safe HVACR and Cold Chain Expo; organised a technical tour to Elite Hitech Industries Ltd.

IMechE BUET Student Chapter

2024 – Present

Chair (Jul 2026 – Present); Treasurer (Apr 2025 – Jul 2026)

BUET, Dhaka

- Directing chapter strategy, programming, and liaison with the Institution of Mechanical Engineers (IMechE) for the 2026–27 session.
- Ran the Speak Out for Engineering (SOE) Intra BUET Heats 2026 end to end as Chair, financed entirely from chapter funds; organised the joint ASHRAE and IMechE stall at BUET Club Fest 2026.
- As Treasurer: managed the 2025–26 budget and a large-scale recruitment drive; ran logistics for two drone technology workshops and masterclasses with Distinguished Prof. Saidur Rahman and Erasmus Mundus representatives.

Multiscale Mechanical Modeling and Research Network (MMMRN)

Jul 2026 – Present

Vice President, Offline Affairs

BUET, Dhaka

- Working to establish MMMRN as an officially recognised BUET club; liaised on a joint session with the AUST Applied Simulation and Modeling Lab on MD simulation with LAMMPS.

BUET Automobile Club

2024 – Present

Secretary (Jul 2026 – Present); Additional Chief Executive, Logistics (2024 – Jul 2026)

BUET, Dhaka

- Contribute to executive decisions; helped organise the freshers' reception, organised the outgoing panel's farewell, and supported logistics for automotive and robotics exhibitions.

Badhan Foundation

2022 – Present

Volunteer, voluntary blood donation organisation

Dhaka, Bangladesh

- Coordinate donors for blood donation drives and medical emergencies.

Teaching and Mentoring

Independent Academic Tutor

2022 – Present

STEM and university admission mentor

Dhaka, Bangladesh

- Mentored 60+ secondary and higher secondary students in science and mathematics; prepared students who secured admission to BUET, RUET, KUET, and public medical colleges.

Certifications

- **Certified SOLIDWORKS Professional (CSWP)**, Dassault Systèmes, Jan 2026 (ID C-CJUFSAP6EX).
- **Certified SOLIDWORKS Associate (CSWA)**, Dassault Systèmes, Nov 2024 (ID C-73JNRWCGL5).
- **SOLIDWORKS Associate** in Simulation, Additive Manufacturing, and Sustainability, Dassault Systèmes, Sep 2025.

Workshops and Training

- Mechanical Course, Team Interplanetar, BUET: Certificate of Excellence for a week-long rover design programme (Jul 2024).
- ANSYS Unveiled: finite element analysis and CFD, BUET Automobile Club (Jul 2023).
- Exploring Mechanical Research: An Introduction to MMMRN, BUET (Jun 2023).
- Drone technology: Lift-Off (Jun 2025) and Shape-shift Drone (2025), IMechE BUET Student Chapter; Hands-on Trash Collector Robot, BUET Robotics Society (Jun 2023).